

7. Ilyashenko, O. Yu. Possibilities of Using Computer Vision for Data Analysis in Medicine / O. Yu. Ilyashenko, E. L. Lukyanchenko // Bulletin of Saratov University. Mathematics. Mechanics. Informatics, 2022. – Vol. 22, No. 2. – Pp. 224-232.
8. Bagrintseva, O. B. Possibilities of Using Artificial Intelligence in English Classes with Students of Engineering Specialties (on the Example of the Field of Study "Agricultural Engineering") / O. B. Bagrintseva, I. A. Grokhovskaya // Pedagogical Research. – 2024. – № 3. – pp. 105-127.
9. Kasatkin, A. A. Development and Application of Machine Vision and Image Processing Algorithms / A. A. Kasatkin, N. I. Limanova, V. V. Kozlov // Trends in Science and Education Development, 2023. – No. 98-10. – Pp. 58-61.
10. Mikhailov, I. V. Agile approach in education: implementation practice / I. V. Mikhailov, O. B. Bagrintseva // November, 09, 2023. – P. 157-159.
11. Manannikov, A. O. Use of Artificial Intelligence Technologies in Medicine: Modern Approaches and Prospects / A. O. Manannikov, L. V. Gaev // Trends in Science and Education Development, 2024. – No. 111-8. – Pp. 58-61.
12. Marushchak, E. A., Zubareva, E. A., Glushkov, P. S., Fisenko, E. P. Evaluation of Artificial Intelligence Application Results in Ultrasound Diagnostics of Breast Tumors // Russian Electronic Journal of Radiological Diagnostics, 2025. – Vol. 15, No. 1. – Pp. 119-129.
13. Ponkin, I. V. Medical Imaging and Computer Vision: Concepts, Problems, Legal Aspects / I. V. Ponkin // Business, Management and Law, 2024. – No. 2(62). – Pp. 52-58.
14. Shaderkina, A. I., Lebedev, G. S., Maslyukov, A. P., Shaderkin, I. A. Deep Machine Learning (Artificial Intelligence) in Ultrasound Diagnostics // Journal of Telemedicine and e-Health, 2020. – Vol. 6, No. 2. – Pp. 22-29.

UDC 004.8

COMPUTER VISION IN MEDICAL IMAGE ANALYSIS: AN OVERVIEW AND KEY DIRECTIONS

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Abstract: *Modern veterinary medicine, like human medicine, is characterized by an exponential growth in the volume and complexity of medical visual data. Diagnostic methods such as computed tomography (CT), magnetic resonance imaging (MRI), radiography, ultrasound examination (ultrasound), and cytology generate huge arrays of information, the interpretation of which requires significant time and high qualifications from doctors. In the context of veterinary practice,*

where one specialist often encounters the anatomical diversity of species from small rodents to large agricultural animals, this task becomes especially difficult. In this context, computer vision (CV) technologies are becoming a key tool that transforms approaches to the analysis of medical images.

Keywords: *Computer Vision, veterinary medicine, medical imaging, diagnostics, diagnostic standardization, radiography, ultrasonography, cytology.*

Computer vision is a field of artificial intelligence that allows machines to «see», interpret, and understand visual information[6]. Applied to medical imaging, its goal is to automate and improve the accuracy of tasks such as detecting pathologies, segmenting organs and tissues, classifying diseases, and supporting diagnostic decision-making. The importance of this technology is difficult to overestimate, as it is revolutionizing diagnostic approaches, transforming passive images into active, intelligent sources of clinically significant information. The main value lies in the ability of algorithms to objectively and quantitatively analyze visual data at a level of complexity and consistency that is often unattainable for human perception [3]. This allows for a transition from an intuitive, experience-based assessment to a precise, measurable, and reproducible analysis, freeing up the clinician's time for making a final diagnosis and making complex tactical decisions [1].

The profound impact of computer vision is manifested in its ability to identify hidden, subclinical patterns invisible to the naked eye. Algorithms, especially those based on deep learning, can detect pathologies on radiographic images, assess subtle changes in the texture of organ parenchyma on ultrasound, or identify atypical cells in cytological smears, acting as a highly sensitive early warning system [5]. This opens the way to a fundamentally new level of early diagnosis, when a disease can be detected at the very initial, most treatable stage. Moreover, the technology makes a decisive contribution to the personalization of medicine, allowing for tracking the dynamics of the disease with unprecedented accuracy. For example, instead of just visually comparing CT images of a dog with osteosarcoma before and after chemotherapy, one can accurately calculate the change in tumor volume, assessing the response to treatment objectively and quantitatively, which provides the oncologist with clear, measurable data for timely therapy correction.

Thus, computer vision goes far beyond simple automation, becoming a catalyst for the transition to more accurate, predictive, and preventive medicine. Its application begins with the stage of automatic data preprocessing, where algorithms perform image alignment, noise reduction, and contrast enhancement, which is especially in demand for methods such as MRI or CT. Then the technologies come into force to solve key diagnostic tasks, among which the central place is occupied by automatic segmentation - the precise selection of the boundaries of the anatomical structures or pathological formations of interest. This allows, with unprecedented accuracy and speed, to measure tumor volume, highlight the contours of the heart to calculate the ejection fraction, or mark the lobes of the liver for surgery planning.

Real cases of the implementation of computer vision in veterinary medicine illustrate its transformative potential. For example, in veterinary orthopedics, deep learning algorithms are successfully used for the automatic diagnosis of hip dysplasia in dogs [8]. A system trained on thousands of radiographic images is capable of not only determining the degree of dysplasia according to the OFA (Orthopedic Foundation for Animals) standard but also measuring Norberg angles and distraction indices, providing unprecedented objectivity and speed of screening in kennels and at exhibitions. Another example is the analysis of thoracic radiographs in cats suspected of having cardiomegaly. The algorithm automatically calculates the Vertebral Heart Score (VHS), highlighting the contours of the heart and vertebrae, which allows for quickly and accurately identifying animals with hypertrophic cardiomyopathy, one of the most common pathologies in this species [2].

In veterinary ultrasonography, computer vision has found application for solving the problem of operator dependence. Thus, during ultrasound of the thyroid gland in dogs, algorithms perform texture analysis of the parenchyma and automatically measure the volume of the lobes, helping to differentiate hyperplasia from neoplasia. In the reproductive biology of agricultural animals, systems based on CV are used for automatic counting and measurement of follicles on the ovaries of cows, which allows for accurately determining the optimal time for insemination and increasing the efficiency of artificial insemination. And in veterinary echocardiography, software systems in real time track the movement of the ventricular walls and automatically calculate the ejection fraction and other hemodynamic parameters, which is critically important for managing patients with heart failure.

Another critically important application is the automatic detection and classification of pathologies. Algorithms trained on extensive data archives have learned to detect deviations, subtle to the human eye, with the highest sensitivity. A striking example from veterinary practice is the use of computer vision for screening tuberculosis in cattle using radiographic images of the lungs [4]. An algorithm developed by one of the European laboratories is able to identify characteristic granulomas and calcifications with an accuracy exceeding 95%, which allows for mass and rapid examination of livestock. Similar systems are used for diagnosing pneumonia in calves and pigs using radiographs, which is especially in demand in large livestock complexes.

The potential of the technology extends even further, into the field of predictive analytics. By analyzing complex patterns and radiomics - the extraction of quantitative characteristics from medical images - algorithms can predict the aggressiveness of a tumor, the likelihood of a response to a specific type of chemotherapy, or the risks of developing diseases in the future, which lays the foundation for truly personalized medicine. In veterinary oncology, systems are already used today that, based on histological sections of a mammary tumor in a dog, not only classify the type of carcinoma but also predict the likelihood of metastasis based on the analysis of nuclear polymorphism and microvessel density.

Considering specific diagnostic methods, it can be noted that the use of computer vision for the analysis of ultrasound images radically increases their information content, accuracy, and efficiency. The key advantage here lies in overcoming the fundamental limitations inherent in ultrasound - namely, high operator dependence and subjectivity of interpretation. Algorithms provide unprecedented objectivity, transforming visual information into precise, measurable metrics. In veterinary practice, this has been reflected in the development of portable ultrasound scanners with built-in AI for farmers. Such devices allow even non-specialists to diagnose pregnancy in cows at early stages: the system automatically recognizes the embryo and assesses its viability, significantly increasing the efficiency of herd reproduction management.

The impact of computer vision is also manifested in its ability to identify subclinical patterns invisible to the human eye in the ultrasound picture. Algorithms can perform complex texture analysis of organ parenchyma, for example, differentiating degrees of liver fibrosis in cats with lipidosis or detecting early signs of chronic pancreatitis in dogs, acting as a powerful early warning system. The technology also radically accelerates work with volumetric data, for example, during echocardiography, where the system can semi-automatically highlight the boundaries of the heart chambers and instantly calculate key parameters of myocardial function.

Similarly, the application of computer vision for the analysis of radiographic images marks a qualitative leap in veterinary radiodiagnostics. The main advantage of the technology lies in its ability to radically improve the accuracy and reliability of diagnostics, acting as an unbiased and tireless expert. Algorithms demonstrate the highest sensitivity in detecting pathologies that may be missed by the human eye, such as micro-fractures in racehorses, early infiltrative changes in the lung tissue during pneumonia in reptiles, or minimal effusions. Computer vision acts as a double-check system, significantly reducing the level of false-negative conclusions. A vivid example is the use of AI algorithms in veterinary clinics for the automatic detection of foreign bodies in the gastrointestinal tract of dogs, which allows for quickly identifying radiopaque objects on abdominal images and reducing the time before surgical care is provided.

Furthermore, the technology brings an unprecedented level of standardization, allowing for the automation of labor-intensive processes, such as performing complex measurements, in a matter of seconds, which not only saves the specialist's time but also eliminates inter-operator variability. The most revolutionary potential of computer vision in veterinary radiology is revealed in its ability for predictive analytics, allowing for the identification of subtle patterns associated with the early stages of diseases, long before the appearance of a clear clinical picture. For example, algorithms, analyzing a series of radiographs of the hip joints in growing puppies of large breeds, can predict the risk of developing severe forms of dysplasia, allowing for the timely initiation of preventive therapy.

Finally, the introduction of computer vision into veterinary cytological diagnostics marks a real revolution, transferring this highly informative but

subjective field to the rails of objective, quantitative analysis [7]. The main advantage here lies in the ability to radically improve the accuracy and reproducibility of the assessment of cellular material. Algorithms demonstrate exceptional sensitivity in detecting and classifying cells, are able to identify atypical or rare cell populations that may be missed during routine examination, acting as the most powerful safety net. In practical veterinary medicine, this is implemented in the form of automated systems for analyzing blood smears, which in just a few minutes perform a differential count of leukocytes and identify pathological cell forms, for example, in lymphoma in cats or babesiosis in dogs. Such systems are already operating in large veterinary diagnostic centers, significantly accelerating diagnosis.

Computer vision introduces an unprecedented level of standardization and depth into morphological analysis, conducting a complex quantitative analysis of thousands of cells in one smear and assessing subtle but clinically significant changes in morphology. This allows not only stating the fact of atypia but also objectively measuring its degree, creating a unique «digital signature» of the tumor. The technology radically speeds up and automates the most routine stages of the cytologist's work, such as the differential count of leukocytes, which not only multiplies the laboratory's throughput but also frees up the specialist's time for analyzing the most complex cases. For example, when analyzing aspirates of mammary tumors in dogs, the system can not only classify the type of neoplasm but also assess nuclear polymorphism and the number of mitoses, providing the veterinarian with a detailed cytological report within minutes.

Ultimately, the integration of computer vision into clinical veterinary practice allows not only automating routine tasks but also obtaining fundamentally new, previously inaccessible diagnostic information. It acts not as a replacement for the veterinarian, but as his intellectual assistant, who takes on routine tasks, provides expert quantitative analysis, and provides data for making a more informed, faster, and accurate diagnostic decision. This increases the efficiency, speed, and, most importantly, the accuracy of veterinary care at all stages - from screening and diagnosis to treatment planning and monitoring its effectiveness, ultimately improving outcomes for patients of all species.

Список литературы:

1. Гроховская, И. А. Уровни коммуникации в границах медицинского дискурса / И. А. Гроховская // Термины в коммуникативном пространстве : материалы всероссийской научно-практической конференции с международным участием, Астрахань, 08 февраля 2024 года. – Астрахань: Астраханский государственный медицинский университет, 2024. – С. 134-139.
2. Анализ причин распространения парвовирусной инфекции собак в условиях Г. Астрахани / Ю. К. Сергиенко, В. В. Зайцев, А. А. Лифанова [и др.] // Прикаспийский международный молодёжный научный форум агропромтехнологий и продовольственной безопасности 2018 : сборник научных статей, Астраханский государственный университет, 19–20 апреля

2018 года. – Астраханский государственный университет: Федеральное государственное бюджетное образовательное учреждение высшего профессионального образования "Астраханский государственный университет", 2018. – С. 67-69.

3. Эффективность применения препарата "Гепасейф" при вирусных гепатитах собак / А. С. Ванина, Н. А. Воронцева, В. В. Зайцев, А. П. Полковниченко // Прикаспийский международный молодежный научный форум агропромтехнологий и продовольственной безопасности 2017 : Сборник научных статей, Астрахань, 20–21 апреля 2017 года / Составители А.Р. Лозовский, О.Н. Беспалова, А.А. Айтпаева, Ю.И. Шахмедова. – Астрахань: Федеральное государственное бюджетное образовательное учреждение высшего профессионального образования "Астраханский государственный университет", 2017. – С. 38-40.

4. Особенности про- и антиоксидантной системы крупного рогатого скота в постнатальном онтогенезе в биогеохимических условиях Астраханской области / И. С. Михайлова, В. В. Зайцев, В. М. Яралиев [и др.] // Ученые записки Казанской государственной академии ветеринарной медицины им. Н.Э. Баумана. – 2021. – Т. 247, № 3. – С. 150-156. – DOI 10.31588/2413-4201-1883-247-3-150-156.

5. Багринцева, О. Б. Возможности применения искусственного интеллекта на занятиях по английскому языку со студентами инженерных специальностей (на примере направления подготовки «Агроинженерия») / О. Б. Багринцева, И. А. Гроховская // Педагогические исследования. – 2024. – № 3. – С. 105-127.

6. Mikhailov, I. V. Agile approach in education: implementation practice / I. V. Mikhailov, O. B. Bagrintseva // 09 ноября 2023 года, 2023. – P. 157-159.

7. Banzato, T., Bonsemblante, F., Aresu, L., Gelain, M. E., & Burti, S. A deep learning approach to the automatic classification of thoracic radiographs in cats // *Veterinary Radiology & Ultrasound*, 2018. – Pp. 513-519.

8. Belli, C. B., Maki, J., & Farnsworth, R. The role of artificial intelligence in veterinary diagnostic imaging // *Veterinary Sciences*, 2019. – Pp. 63-69.

9. Broughan, J. M., Judge, J., Ely, E., Delahay, R. J., Wilson, G., Clifton-Hadley, R. S., Downs, S. H. A review of risk factors for bovine tuberculosis in cattle // *Veterinary Journal*, 2016. – Pp. 22-29.

10. Esteva, A., Chou, K., Yeung, S., Naik, N., Madani, A., Mottaghi, A., Socher, R. Deep learning-enabled medical computer vision // *NPJ Digital Medicine*, 2021. – Pp. 1-9.

11. Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., Sánchez, C. I. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 2017. – Pp. 60-88.

12. Oliveira, L. P., & de Almeida, V. M. Machine learning in veterinary cytology: a review. // *Veterinary Clinical Pathology*, 2020. – Pp. 35-44.

13. Porrello, A., Bonicelli, L., Arcelli, C., Tassani, S., & Bonaldo, A. Automatic hip dysplasia assessment from radiographs using deep learning. *IEEE Journal of Biomedical and Health Informatics*, 2021. – Pp. 3127-3135.

References:

1. Grokhovskaia, I. A. Communication levels within the boundaries of medical discourse / I. A. Grokhovskaia // Terms in the communicative space: materials of the All-Russian scientific and practical conference with international participation, Astrakhan, the 8th of February, 2024. – Astrakhan: Astrakhan State Medical University, 2024. – Pp. 134-139.
2. Sergienko, Yu. K. Analysis of the Causes of the Spread of Canine Parvovirus Infection in the Conditions of Astrakhan / Yu. K. Sergienko, V. V. Zaitsev, A. A. Lifanova [et al.] // Caspian International Youth Scientific Forum of Agrotechnologies and Food Security 2018 : collection of scientific articles, Astrakhan State University, April 19–20, 2018. – Astrakhan State University: Federal State Budgetary Educational Institution of Higher Professional Education "Astrakhan State University", 2018. – pp. 67-69.
3. Vanina, A. S. Effectiveness of the Drug "Gepasafe" in Viral Hepatitis of Dogs / A. S. Vanina, N. A. Vorontseva, V. V. Zaitsev, A. P. Polkovnichenko // Caspian International Youth Scientific Forum of Agrotechnologies and Food Security 2017 : Collection of Scientific Articles, Astrakhan, April 20–21, 2017 / Compiled by A. R. Lozovsky, O. N. Bessalova, A. A. Aitpaeva, Yu. I. Shakhmedova. – Astrakhan: Federal State Budgetary Educational Institution of Higher Professional Education "Astrakhan State University", 2017. – pp. 38-40.
4. Mikhailova, I. S. Features of the Pro- and Antioxidant System of Cattle in Postnatal Ontogenesis under Biogeochemical Conditions of the Astrakhan Region / I. S. Mikhailova, V. V. Zaitsev, V. M. Yarialiev [et al.] // Scientific Notes of the Kazan State Academy of Veterinary Medicine named after N. E. Bauman. – 2021. – Vol. 247, No. 3. – pp. 150-156. – DOI 10.31588/2413-4201-1883-247-3-150-156.
5. Bagrintseva, O. B. Possibilities of Using Artificial Intelligence in English Classes with Students of Engineering Specialties (on the Example of the Field of Study "Agricultural Engineering") / O. B. Bagrintseva, I. A. Grokhovskaya // Pedagogical Research. – 2024. – № 3. – pp. 105-127.
6. Mikhailov, I. V. Agile approach in education: implementation practice / I. V. Mikhailov, O. B. Bagrintseva // November, 09, 2023. – P. 157-159.
7. Banzato, T., Bonsembiante, F., Aresu, L., Gelain, M. E., & Burti, S. A deep learning approach to the automatic classification of thoracic radiographs in cats // Veterinary Radiology & Ultrasound, 2018. – Pp. 513-519.
8. Belli, C. B., Maki, J., & Farnsworth, R. The role of artificial intelligence in veterinary diagnostic imaging // Veterinary Sciences, 2019. – Pp. 63-69.
9. Broughan, J. M., Judge, J., Ely, E., Delahay, R. J., Wilson, G., Clifton-Hadley, R. S., Downs, S. H. A review of risk factors for bovine tuberculosis in cattle // Veterinary Journal, 2016. – Pp. 22-29.
10. Esteva, A., Chou, K., Yeung, S., Naik, N., Madani, A., Mottaghi, A., Socher, R. Deep learning-enabled medical computer vision // NPJ Digital Medicine, 2021. – Pp. 1-9.

11. Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., Sánchez, C. I. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 2017. – Pp. 60-88.
12. Oliveira, L. P., & de Almeida, V. M. Machine learning in veterinary cytology: a review. // *Veterinary Clinical Pathology*, 2020. – Pp. 35-44.
13. Porrello, A., Bonicelli, L., Arcelli, C., Tassani, S., & Bonaldo, A. Automatic hip dysplasia assessment from radiographs using deep learning. *IEEE Journal of Biomedical and Health Informatics*, 2021. – Pp. 3127-3135.

UDC 004.8

AI FOR ACCESSIBILITY: HELPING PEOPLE WITH DISABILITIES THROUGH SMART TECHNOLOGIES

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Abstract: *Artificial Intelligence (AI) is transforming accessibility for people with disabilities by providing innovative solutions that enhance independence, communication, and quality of life. AI-powered technologies, such as speech recognition, computer vision, predictive text, and assistive robotics, enable individuals with physical, sensory, and cognitive impairments to overcome barriers in daily activities. This paper explores the role of AI in improving accessibility, examines current applications, and discusses challenges and ethical considerations. The study highlights how AI technologies can foster inclusive societies and empower people with disabilities to participate fully in social, educational, and professional environments.*

Keywords: *Artificial Intelligence, Accessibility, Disabilities, Assistive Technologies, Inclusive Design, Smart Technologies.*

Accessibility remains a critical challenge for millions of people worldwide living with disabilities [1]. Physical, sensory, and cognitive impairments often create barriers that limit participation in education, employment, communication, and daily life. Traditional assistive technologies, while helpful, have limitations in adaptability, personalization, and real-time responsiveness [2]. The advent of Artificial Intelligence (AI) offers new opportunities to enhance accessibility through smart, adaptive, and intelligent systems.